UNIVERSIDAD NACIONAL DE COLOMBIA

Respuestas - Taller de Calculo Integral I

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1 Integrales Indefinidas Basicas

1. $\int (6x^5 - 4x^2 + 7)\,dx = x^6 - \frac{4}{3}x^3 + 7x + C.$
2. $\int \left(\frac{5}{x} - 3\sqrt{x} + \frac{2}{x^2}\right)\,dx = 5\ln|x| - 2x^{3/2} - \frac{2}{x} + C.$
3. $\int (4e^{2x} - 3\cos x)\,dx = 2e^{2x} - 3\sin x + C.$
4. $\int (\sec^2 x + 5\csc x \cot x)\,dx = \tan x - 5\csc x + C.$
5. $\int (2x+1)^7\,dx = \frac{(2x+1)^8}{16} + C.$
6. $\int \frac{3x^2+1}{x^3+x+4}\,dx = \ln|x^3 + x + 4| + C.$

2 Sustitucion

1. $\int x\sqrt{x^2+9}\,dx = \frac{1}{3}(x^2+9)^{3/2} + C.$
2. $\int \frac{2x-1}{x^2-x+5}\,dx = \ln(x^2 - x + 5) + C.$
3. $\int \frac{\cos(3x)}{1+\sin(3x)}\,dx = \frac{1}{3}\ln|1 + \sin(3x)| + C.$
4. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\,dx = 2e^{\sqrt{x}} + C.$
5. $\int \frac{1}{x\ln x}\,dx = \ln(\ln x) + C,\ x > 1.$

3 Integrales Trigonometricas Elementales

1. $\int \sin(4x)\,dx = -\frac{1}{4}\cos(4x) + C.$
2. $\int \cos^3 x \sin x\,dx = -\frac{\cos^4 x}{4} + C.$
3. $\int \tan x\,dx = \ln|\sec x| + C.$
4. $\int \sec x(\sec x + \tan x)\,dx = \tan x + \sec x + C.$
5. $\int \frac{dx}{1+\cos x} = \tan\left(\frac{x}{2}\right) + C.$

4 Integrales Definidas Basicas

1. $\int_0^2 (3x^2 - 4x + 1)\,dx = 2$.
2. $\int_{-1}^1 (x^3 + 2x^2)\,dx = \frac{4}{3}$.
3. $\int_0^{\pi/2} \sin x\,dx = 1$.
4. $\int_1^e \frac{1}{x}\,dx = 1$.
5. $\int_0^1 \frac{2x}{x^2+1}\,dx = \ln 2$.

5 Aplicaciones Cortas

1. Si $f'(x) = 6x^2 - 4x + 3$, entonces $f(x) = 2x^3 - 2x^2 + 3x + C$. Con $f(1) = 5$ se obtiene $C = 2$. Por tanto,

$$f(x) = 2x^3 - 2x^2 + 3x + 2.$$

2. Area bajo $y = 2x + 1$ entre 0 y 4:

$$A = \int_0^4 (2x + 1)\,dx = [x^2 + x]_0^4 = 20.$$

3. Area bajo $y = x^2$ en $[0, 3]$:

$$A = \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = 9.$$